Microsoft Elevate

CAPSTONE PROJECT

Blinkit Grocery Sales Data Analysis

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OUTLINE:

- Problem Statement
- Proposed System/Solution
- System Development Approach
- Algorithm & Deployment
- Result (Output Image)
- Conclusion
- Future Scope
- References

PROBLEM STATEMENT:

•Urban quick-commerce platforms like Blinkit face challenges in managing grocery sales efficiently.
•Businesses often lack insights into customer purchasing patterns and outlet performance.
•Without data-driven analysis, decision-making relies on assumptions, leading to:
•Overstock or stockouts of grocery items
•Inefficient allocation of resources across outlets
•Loss of potential revenue
•Decreased customer satisfaction
•There is a need to **analyze sales data, customer ratings, and outlet details** to identify trends and optimize operations.
•Goal: Enable stakeholders to make **strategic decisions backed by actionable insights**.

PROPOSED SOLUTION:

- Use historical Blinkit sales data to **understand sales patterns and customer preferences**.
- Analyze outlet performance based on **type, size, location, and age**.
- Integrate multiple tools for end-to-end analysis:
- **SQL** – Data extraction, transformation, aggregation
- **Python** – Data cleaning, EDA, visualizations
- **Excel** – Quick KPI verification
- **Power BI** – Interactive dashboards for business insights
- Key outcomes:
- Identify high- and low-performing products and outlets
- Suggest strategies to improve revenue and efficiency
- Provide a **visual, interactive, and actionable dashboard** for stakeholders

SYSTEM APPROACH:

Step 1: Data Collection

•Gather sales, outlet, and customer rating data

•Include product details: item type, fat content, and revenue generated

Step 2: Data Preprocessing

•Clean missing values, remove duplicates

•Feature engineering for metrics like avg. sales, avg. rating

Step 3: Analysis & Visualization

•Python: Grouped metrics, trends, and correlation analysis

•Power BI: Interactive charts, matrices, KPI cards

Step 4: Dashboard Deployment

•Real-time insights for stakeholders

•Drill-down, dynamic filtering, and geo-based analysis

ALGORITHM & DEPLOYMENT:

•**SQL**: Extract and aggregate sales, ratings, and outlet data
•**Python (Pandas, Matplotlib, Seaborn)**:
•Data cleaning and preprocessing
•Exploratory Data Analysis (EDA)
•Trend visualization and correlation study
•**Power BI**:
•Interactive dashboard with charts (donut, bar, stacked column, funnel map, KPI cards)
•Dynamic filters and drill-downs for stakeholder interaction
•**Workflow Summary:**
1.Data extraction → 2. Data cleaning → 3. EDA → 4. KPI calculation → 5. Dashboard visualization

RESULT:

Sales Analysis:

•Top-performing product categories contributing most to revenue

•Revenue distribution by fat content, item type, outlet type, and location

Outlet Performance:

•Larger and older outlets in premium locations generated higher revenue

Customer Behavior:

•High ratings correlated with best-selling products and outlets

•Trends in purchases by time, location, and category

drill-down analysis

CONCLUSION:

- The project **demonstrates the power of data-driven decision making** in retail grocery.
- Stakeholders can **identify trends, optimize inventory, and improve outlet performance**.
- Integration of **Python, SQL, Excel, and Power BI** provides a full business intelligence workflow.
- The interactive dashboard allows stakeholders to **filter, drill-down, and view insights quickly**.
- Outcome: Improved revenue potential, efficient operations, and better customer satisfaction.

FUTURE SCOPE:

- Integrate **real-time sales data** for live monitoring
- Add **predictive analytics** for inventory demand forecasting
- Include **marketing analytics**: promotions, discounts, campaigns
- Expand analysis to **multiple cities or regions** for a broader business strategy
- Incorporate **sentiment analysis** from customer feedback for deeper insights

REFERENCES:

Demand Prediction on Bike Sharing Data Using Regression Analysis – A study comparing multiple statistical models for predicting bike demand with weather and time variables, measuring accuracy with RMSE and R-Squared.

Bike-Sharing Demand Forecasting Using Machine Learning – Describes a practical ML pipeline for station-level demand forecasting and deployment in a real system, highlighting the value of ML in operational predictions.

GitHub Link: https://github.com/gamana29/Blinkit_Data_Analysis.git

Thank You